

Experiment-6

Aim

To simulate the **Sliding Window Protocol** for data transmission.

Procedure

- Start the program.
- Input the **window size (w)**.
- Enter the **number of frames (f)** to be transmitted.
- Input the sequence of frames.
- The sender transmits frames in groups equal to the window size.
- After sending each window, the sender waits for acknowledgement.
- Continue until all frames are transmitted.

Output

```

main.c [c -] - CodeBlocks 20.03
File Edit View Search Project Build Debug Runman wsSmith Tools Tools- Plugins DoxyBlocks Settings Help
C:\Users\Tcs\Documents\CodeBlocks\Projects\c.pro\c.m\bin\Debug\c.m.exe
Enter window size: 3
Enter number of frames to transmit: 5
Enter 5 frames: 1 2 3 4 5
<global>With sliding window protocol the frames will be sent in the following manner (assuming no corruption of frames)
After sending 3 frames at each stage sender waits for acknowledgement sent by the receiver
1 2 3
Acknowledgement of above frames sent is received by sender
4 5
Acknowledgement of above frames sent is received by sender
Process returned 0 (0x0)   execution time : 17.403 s
Press any key to continue.
Checking for existence: C:\Users\Tcs\Documents\CodeBlocks\Projects\c.pro\c.m\bin\Debug\c.m.exe
Set variable - PATH: C:\Program Files\CodeBlocks\MinGW\bin;C:\Program Files\CodeBlocks\MinGW\bin\gcc.exe\app\gcc.exe\product\11.2.0\server\bin;C:\ProgramData
\Oracle\Java\javapath;C:\Windows\System32;C:\Windows\System32\wbem;C:\Windows\System32\WindowsPowerShell\v1.0;C:\Windows\System32\OpenSSH;C:\MinGW
\bin;C:\msys64\mingw64\bin;C:\Program Files (x86)\PuTTY\bin;C:\Program Files\Java\jdk1.8.0_40\bin;C:\Program Files\Git\cmd;C:\Users\Tcs\AppData\Local\Programs
\Python\Python812\Scripts;C:\Users\Tcs\AppData\Local\Programs\Python\Python812;C:\Users\Tcs\AppData\Local\Microsoft\WindowsApps\;C:\Users\Tcs\AppData\Local
\Programs\Microsoft VS Code\bin;C:\Program Files (x86)\Unmap
Executing: "C:\Program Files\CodeBlocks\cb_console_runner.exe" "C:\Users\Tcs\Documents\CodeBlocks\Projects\c.pro\c.m\bin\Debug\c.m.exe" (in C:\Users\Tcs
\Documents\CodeBlocks\Projects\c.pro\c.m\.)

```

Observation

- Frames are sent in groups (window size)
- Acknowledgment comes after each group
- Sender waits only after a full window
- Remaining frames sent if not a multiple
- Improves efficiency over one-by-one transmission

Result

Thus, Sliding Window Protocol is successfully simulated with efficient frame transmission using grouped acknowledgements.